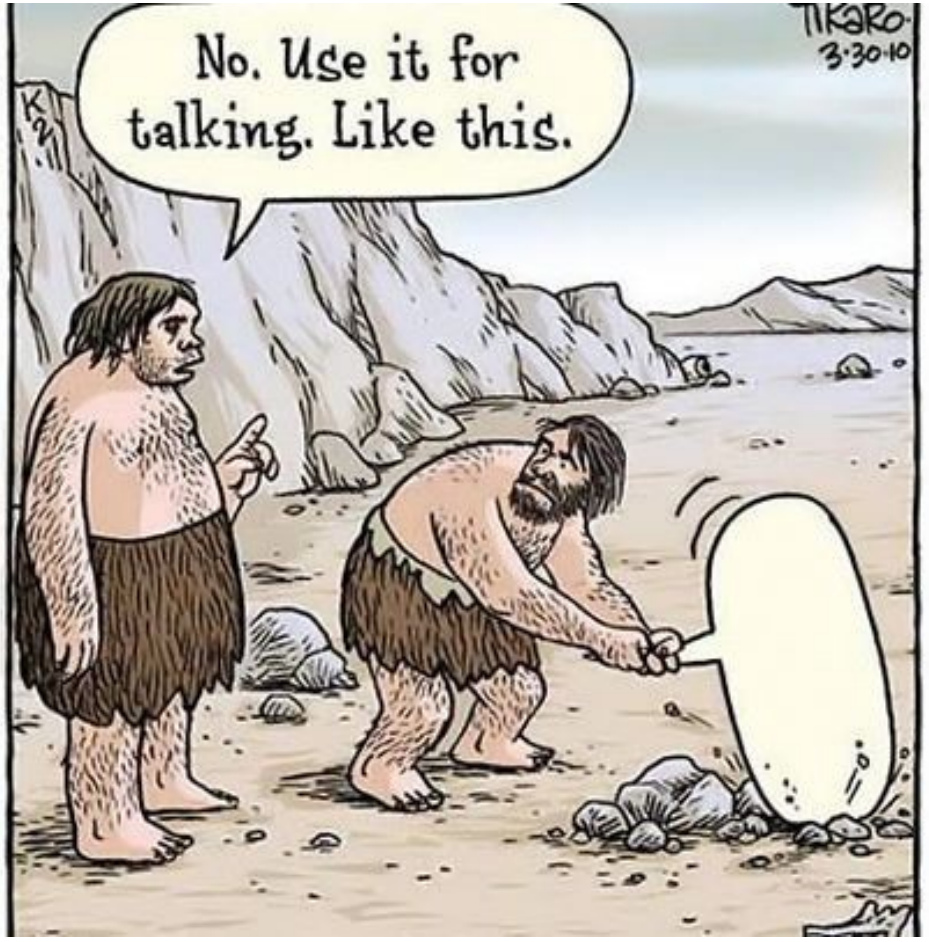
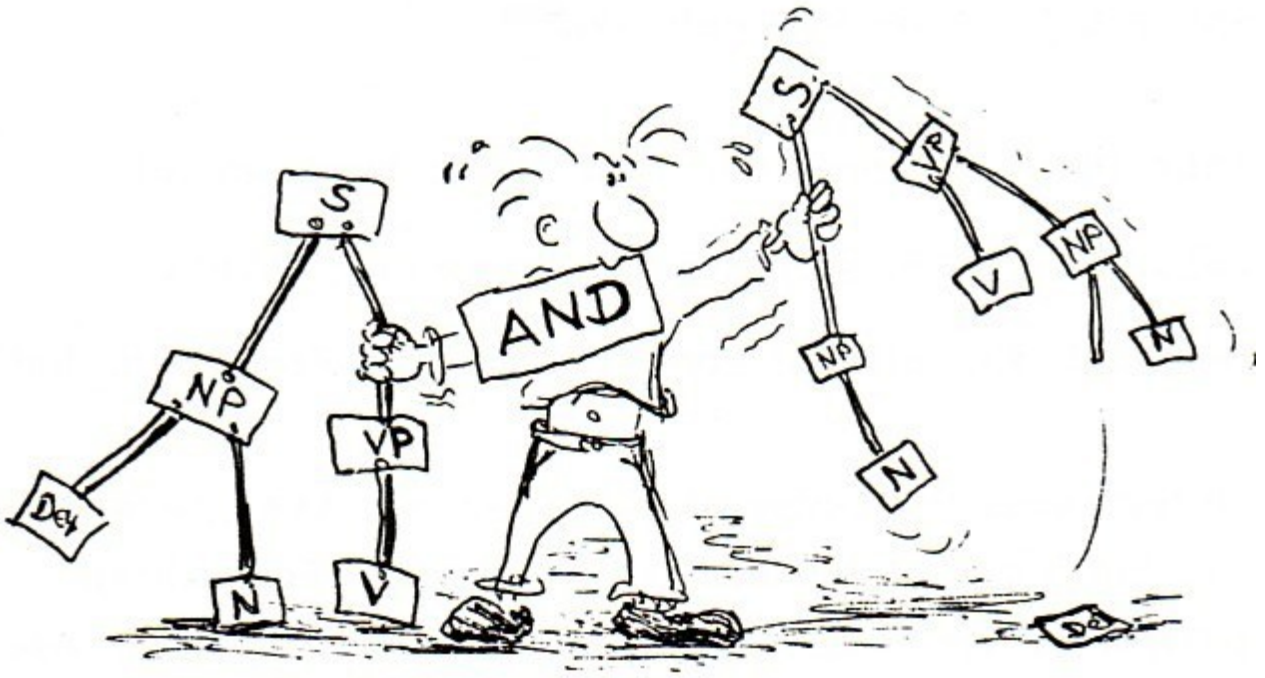
Natural Language Processing

Sentence Syntax



NLP: Syntax (Sentence Level)



Sentence breaking

Sentence boundary disambiguation (SBD), also known as sentence breaking or sentence boundary detection, is the problem in natural language processing of deciding where sentences begin and end. Often, natural language processing tools require their input to be divided into sentences for a number of reasons; however, sentence boundary identification is challenging because punctuation marks are often ambiguous. For example, a period may denote an abbreviation, decimal point, an ellipsis, or an email address – not the end of a sentence. About 47% of the periods in the Wall Street Journal corpus denote abbreviations. As well, question marks and exclamation marks may appear in embedded quotations, emoticons, computer code, and slang.

Languages like Japanese and Chinese have unambiguous sentence-ending markers.

Sentence breaking

The standard approach to locate the end of a sentence:

(a) If it's a period, it ends a sentence.

(b) If the preceding token is in the hand-compiled list of abbreviations, then it doesn't end a sentence.

(c) If the next token is capitalized, then it ends a sentence.

This strategy gets about 95% of sentences correct.

Part of Speech Tagging

Part-of-speech tagging (POS tagging or PoS tagging or POST), also called grammatical tagging or word-category disambiguation, is the process of marking up a word in a text as corresponding to a particular part of speech, based on both its definition and its context—i.e., its relationship with adjacent and related words in a phrase, sentence, or paragraph.

Example: http://parts-of-speech.info/

Example: https://corenlp.run/

Part of Speech Tagging

1. CC Coordinating conjunction
2. CD Cardinal number
3. DT Determiner
4. EX Existential there
5. FW Foreign word
6. IN Preposition or subordinating conjunction
7. JJ Adjective
8. JJR Adjective, comparative
9. JJS Adjective, superlative
10. LS List item marker
11. MD Modal
12. NN Noun, singular or mass
13. NNS Noun, plural
14. NNP Proper noun, singular
15. NNPS Proper noun, plural
16. PDT Predeterminer
17. POS Possessive ending
18. PRP Personal pronoun
19. PRP$ Possessive pronoun
20. RB Adverb
21. RBR Adverb, comparative
22. RBS Adverb, superlative
23. RP Particle
24. SYM Symbol
25. TO to
26. UH Interjection
27. VB Verb, base form
28. VBD Verb, past tense
29. VBG Verb, gerund or present participle
30. VBN Verb, past participle
31. VBP Verb, non-3rd person singular present
32. VBZ Verb, 3rd person singular present
33. WDT Wh-determiner
34. WP Wh-pronoun
35. WP$ Possessive wh-pronoun
36. WRB Wh-adverb

Parsing

Within computational linguistics, the term parsing is used to refer to the formal analysis by a computer of a sentence or other string of words into its constituents, resulting in a parse tree showing their syntactic relation to each other, which may also contain semantic and other information.

Parsing is one of the possible ways to address the PoS problem.

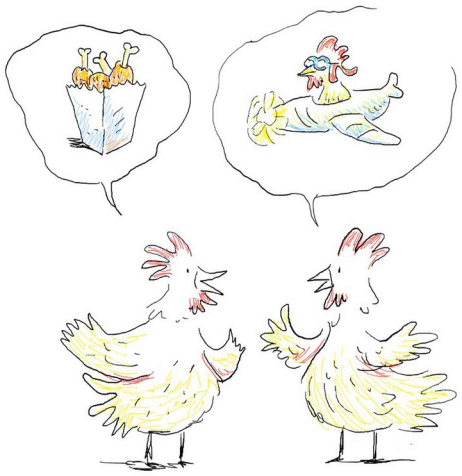
Parsing

Chomsky (1957): Syntactic Structures

Theory of generative grammar (a set of rules that recursively “specify” or “generate” the well-formed expressions of a natural language), considered to be one of the most significant contributions to the field of theoretical linguistics made in the 20th century.



Parsing



Do animals have language?

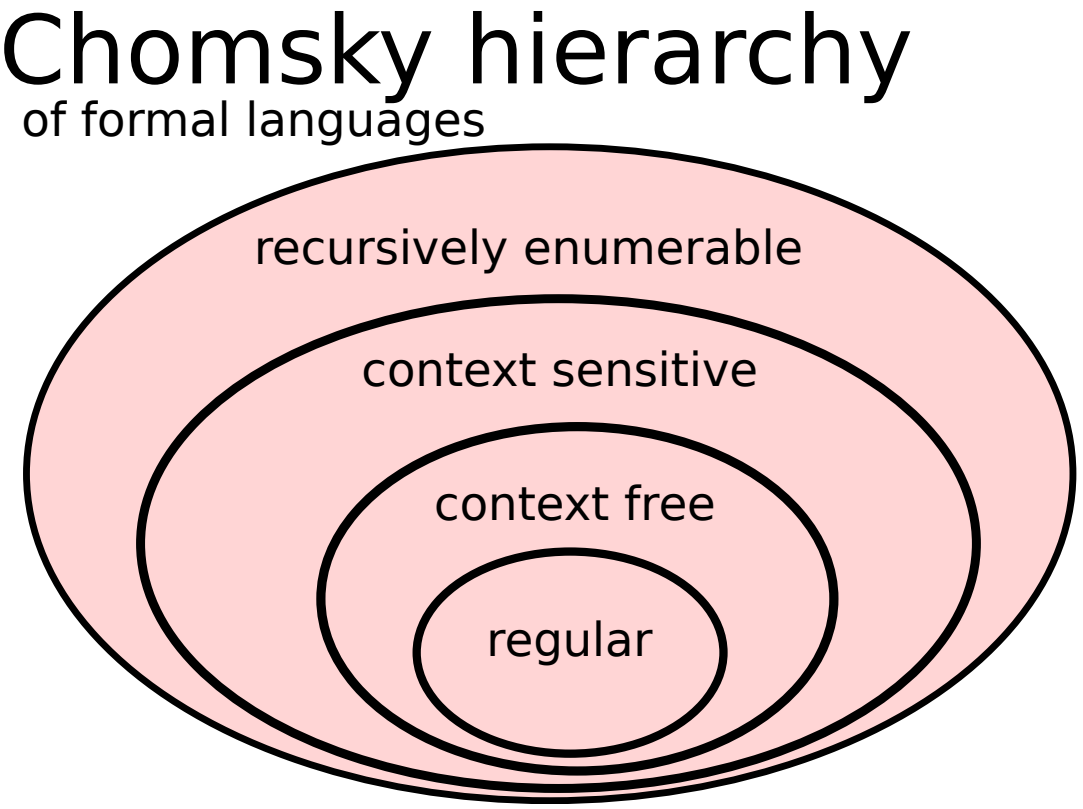


Animal languages are forms of non-human animal communication that show similarities to human language. Animals communicate by using a variety of signs such as sounds or movements. Such signing may be considered complex enough to be called a form of language if the inventory of signs is large, the signs are relatively arbitrary, and the animals seem to produce them with a degree of volition.

While the term "animal language" is widely used, researchers agree that animal languages are not as complex or expressive as human language. Many researchers argue that animal communication lacks a key aspect of human language, that is, the creation of new patterns of signs under varied circumstances.

Parsing

Establishment of the Chomsky – Schützenberger hierarchy, a classification of formal languages in terms of their generative power.



Parsing

The Chomsky Hierarchy

Gram mar	Languages	Automaton	Production rules
Type-0	Recursively enumerable	Turing machine	No restrictions
Type-1	Context-sensitive	Linear-bounded non-deterministic Turing machine	$\alpha A \beta \rightarrow \alpha \gamma \beta$
Type-2	Context-free	Non-deterministic pushdown automaton	$A \rightarrow \gamma$
Type-3	Regular	Finite state automaton	$A \rightarrow aB$ $A \rightarrow a$

Parsing

Grammar

Grammar specifies the compositional structure of complex messages
A formal language is a set of strings of terminal symbols
Each string in the language can be analyzed/generated by the grammar
The grammar is a set of rewrite rules, e.g.,

$$S \rightarrow NP\ VP$$
$$Article \rightarrow \textbf{the} \mid \textbf{a} \mid \textbf{an} \mid \ldots$$

Here S is the sentence symbol, NP and VP are nonterminals

Dibris

Parsing

Grammar types

Regular: *nonterminal* $\rightarrow$ **terminal**[*nonterminal*]

$$S \rightarrow \mathbf{a}S$$
$$S \rightarrow \Lambda$$

Context-free: *nonterminal* $\rightarrow$ *anything*

$$S \rightarrow \mathbf{a}S\mathbf{b}$$

Context-sensitive: more nonterminals on left-hand side

$$ASB \rightarrow AA\mathbf{a}BB$$

Recursively enumerable: no constraints

Natural languages probably context-free, parsable in real time!

Dibris

Parsing

Wumpus lexicon

Noun	→	**stench** | **breeze** | **glitter** | **nothing**
		| **wumpus** | **pit** | **pits** | **gold** | **east** | ...
Verb	→	**is** | **see** | **smell** | **shoot** | **feel** | **stinks**
		| **go** | **grab** | **carry** | **kill** | **turn** | ...
Adjective	→	**right** | **left** | **east** | **south** | **back** | **smelly** | ...
Adverb	→	**here** | **there** | **nearby** | **ahead**
		| **right** | **left** | **east** | **south** | **back** | ...
Pronoun	→	**me** | **you** | **I** | **it** | **S/HE** | **Y'ALL** ...
Name	→	**John** | **Mary** | **Boston** | **UCB** | **PAJC** | ...
Article	→	**the** | **a** | **an** | ...
Preposition	→	**to** | **in** | **on** | **near** | ...
RelPronoun	→	**that** | **which** | ...

Divided into closed and open classes

Dibris

Parsing

Wumpus grammar

S	→	*NP VP*	I + feel a breeze
Subject	→	*Pronoun*	I
	|	*Noun*	pits
	|	*Article Noun*	the + wumpus
	|	*Article Adjective Noun*	the + smelly + wumpus
NP	→	*Subject*	the wumpus
	|	*Subject PP*	the wumpus + to the east
	|	*Subject RelClause*	the wumpus + that is smelly
VP	→	*Verb*	stinks
	|	*Verb NP*	feel + a breeze
	|	*Verb Adjective*	is + smelly
	|	*Verb PP*	turn + to the east
	|	*Verb Adverb*	go + ahead
PP	→	*Preposition NP*	to + the east
RelClause	→	*RelPronoun VP*	that + is smelly

Dibris

Parsing

Definite Clause Grammars

- Represent context-free grammars...
- ...but if we exploit arguments (compare with “attribute grammars”) we can express also context-dependent constraints, making the formalism extremely powerful
- natively integrated with the Prolog programming language: a lot of sophisticated reasoning could be added on top of our grammar generator/recognizer (but – with your relief – we will not!)
- https://swish.swi-prolog.org/

Parsing



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